QC STORY PROJECT

Project Title: To eliminate the concern of rocker panel scratches and damages from Marazzo model .

Mahindra and Mahindra Ltd,NASIK_ Plant, _Multimodel TCF.

	Plan	Actual
Start Date	04.05.18	01.10.18
Completion Date	12.05.18	05.11.18

Current Status: **Completed**

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summaries & Plan future actions
- Guidelines to user

Team information

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	BURHAN MERCHANT	L8-O	CAI	8657327872	Merchant.burhan2@mahindra.com
2	Mentor	Taj Shaikh	L5-M	STCF	8378991344	Shaikh.taj@mahindra.com
3	Member 1	AHER NITIN	L6-M	MFG.	9960800220	Aher.nitin@mahindra.com
4	Member 2	SATISH PAGAR	L8-O	MQA	992220953	pagar.satish@mahindra.com
5	Member 3	SACHIN BHAMARE	L7-O	MQA	9158960111	Bhamare.sachin@mahindra.com
6	Member 4	GANESH BENDALE	L7-O	MFG	9881478840	Bendale.ganesh@mahindra.com
7	Member 5	TARIT NANDI	L7-O	MFG	897576608	Nandi.tarit@mahindra.com

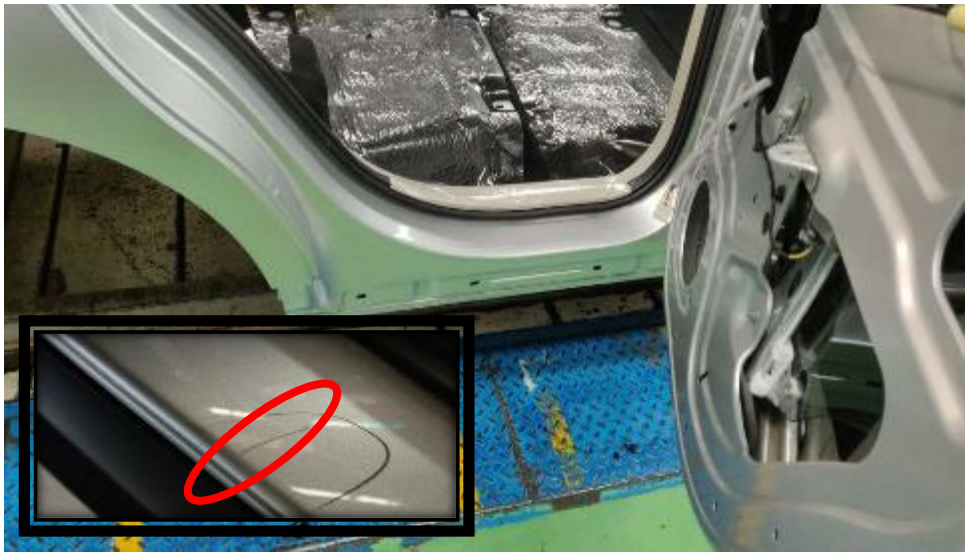
Step1. Select Subject

Brief Description : Front door water leakage in CAI from April 18.

- What : Rocker panel Scratches and damages from Marazzo model .
- Where : In CAI at tester line dent & paint inspection stage.
- When : From June -18.
- Who : By stage associate.
- Why : Rework time high , Affect on CAI WIP, VI severity defect.
- How : 40-50 incidences /per month

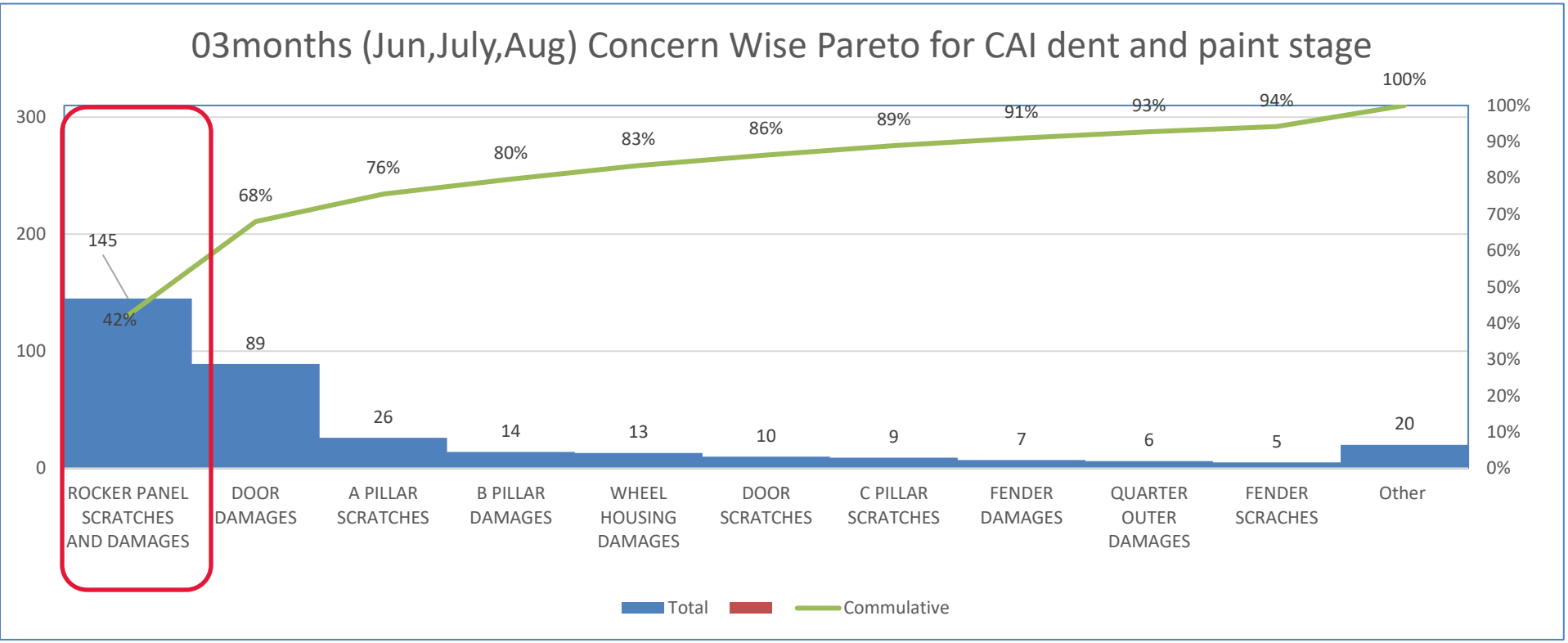
Below are areas where scratches and damges observed :

Rocker panel scratches and damages observed on both i.e., LH /RH side .



2. Justify the choice (reasons)

Pareto analysis of Dent stage Data :-



Inference – Rocker panel scratches and damages is the top contributing concern for Dent and paint DPT.

2. Justify the choice (reasons)

Mahindra

PROMISE 2019
Automotive Business

We will…

Double our revenue to ₹53,000 cr (USD 8 bn), while delivering superior financial performance

- Be the 2nd largest OEM in India by volume
- Double the overseas volume with significant presence in 10 markets

By being the most customer centric company

Brand and Customer Satisfaction
- Be among the Top 2 in Customer Satisfaction
- Be among the Top 3 in brand ranking in PV and #1 in UV & SCV
- Have 10 brands with CaPS > 50

Product and Technology
- Launch 18 products
- Be a pioneer in digital customer experience
- Develop one new disruptive business model

Crusade 2.0
- Drive brand preference through Crusade 2.0 of delivering ETR, FTR and DCE

People
- Be the #1 Great Place to Work in the manufacturing sector

Environment
- Reduce carbon and water footprint by 25%

Thus enabling our customers and stakeholders to Rise, as we progress towards being a Top 10 Global Automotive brand.

Mahindra

PROMISE 2019
Nashik Plant 1, Auto Division

We will…

- Produce 230k vehicles in a year as per market requirement
- Fulfill 100% IO requirement for doubling the overseas volume
- Maintain cost leadership in Auto Sector by reducing revenue cost of 5% YoY
- Improve IQS for Nashik models by 25%, ranking in top 3
- Achieve CaPS > 50 for all Nashik models
- Launch 7 new products on time
- Achieve defined Crusade targets through FTR, ETR and DCE
- Achieve operational excelle[illegible] through digitization
- Achieve No.1 status for em[illegible]ee engagement in the Auto Industry
- Make officers future ready [illegible]plement new technologies coming in the next 5-10 ye[illegible]
- Reduce carbon & water foo[illegible]nt by 25%

we
nd.

Departmental Objective –

First Time Right Launch of New Product - Marazzo

2. Justify the choice

D. COPQ (Cost of Poor Quality) Calculation

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	60
Avg Concerns/Day	Nos	2
Total Required Rework time per day	Min	120
Effective working time	Min	465
Operator required /day	nos	0.26
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	83.87
Avg no. of working days	days	24
Operator cost /month	Rs	2012.90
Part reject/month	Nos	500.00
Part Cost	Rs	1.00
Operator Cost/year	**Rs**	**30154.84**

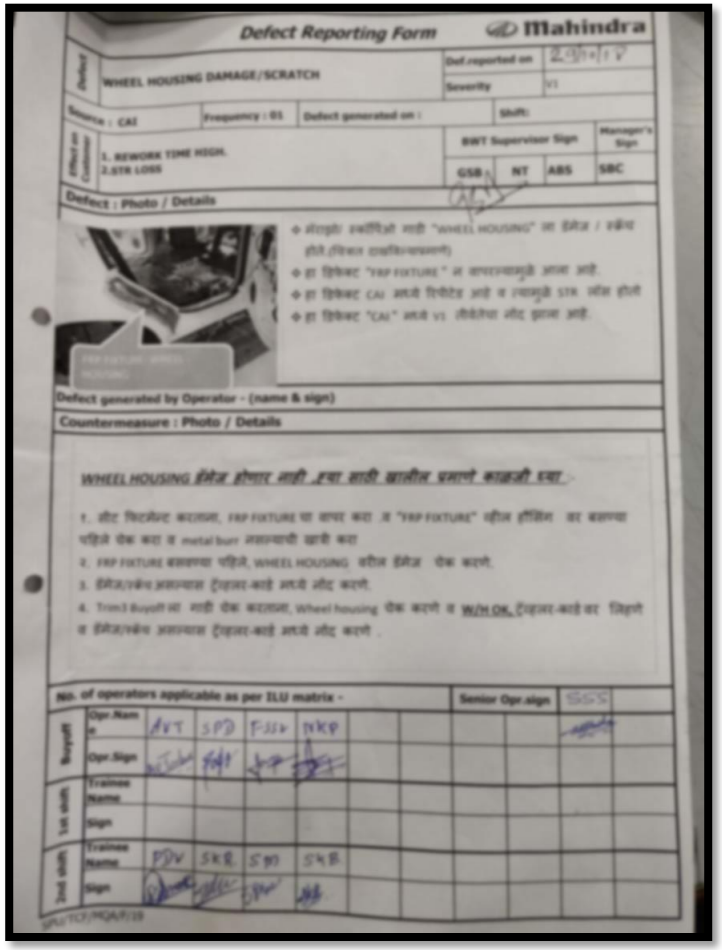
2. Justify the choice (reasons)

Problem Criticality (Severity)

- Rocker panel damage concern contributes to 2% CAI STR loss.
- It creates waiting time to next stage operation.
- Require 60 mins for rework of one vehicle including paint .
- Rocker panel scratches and damages need to bare metal open to repair this concern.
- Rework cost is 30154 Rs.

2.b Immediate Containment Action

- Defect communicated to inspector through DRF .
- TCF all related stage associate awareness given
- Buyoff check sheet modified to detect the source



Defect Reporting Form — Mahindra

Defect: WHEEL HOUSING DAMAGE/SCRATCH

Defect reported on: 2.5/01/17

Severity: V1

Source: CAI | Frequency: 01 | Defect generated on: | Shift:

Effect on Customer: 1. REWORK TIME HIGH. 2.STR LOSS

BWT Supervisor Sign: GSB | NT | ABS | SBC

Manager's Sign

Defect : Photo / Details

Defect generated by Operator - (name & sign)

Countermeasure : Photo / Details

WHEEL HOUSING

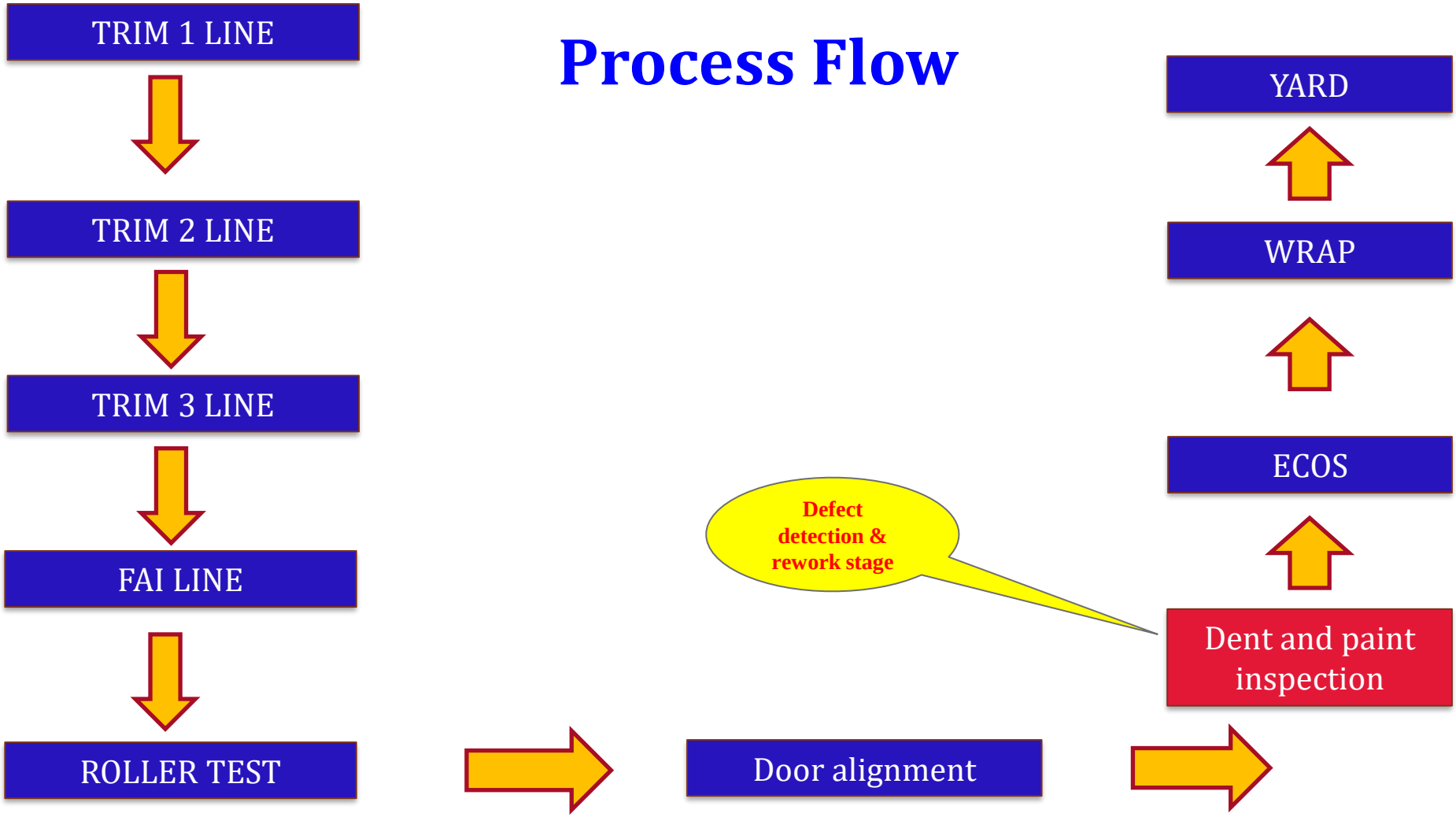
No. of operators applicable as per ILU matrix -

Senior Opr.sign

Buyoff	Opr.Name	AVT	SPD	FJJV	NKP					
	Opr.Sign									
1st shift	Trainee Name									
	Sign									
2nd shift	Trainee Name	PDV	SKB	SPD	SKB					
	Sign									

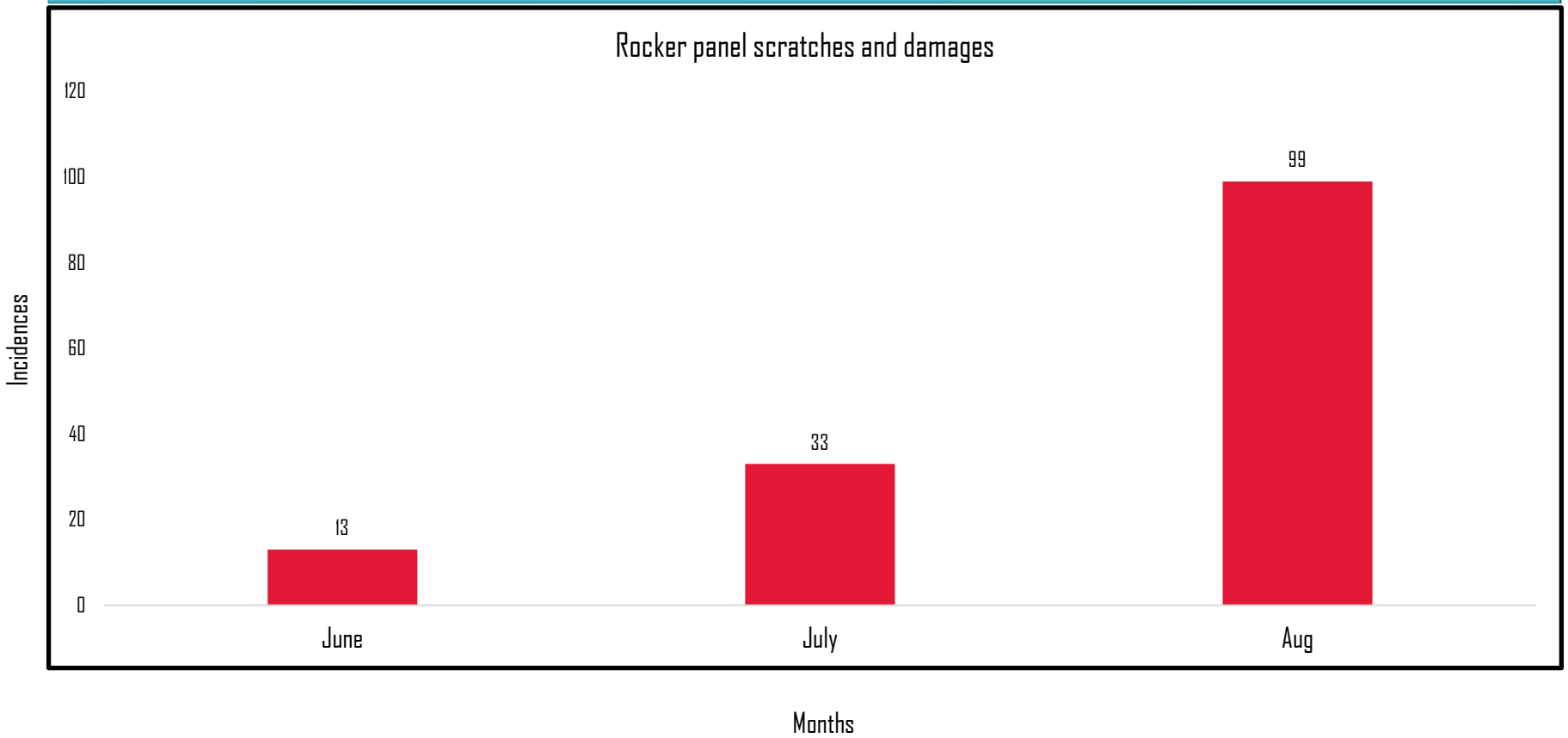
Step 3. Understanding Current Situation

Process Flow



3. Understand the current situation

Month wise **Rocker panel scratches and damages** concern Trend at Dent stage for June to Aug 18



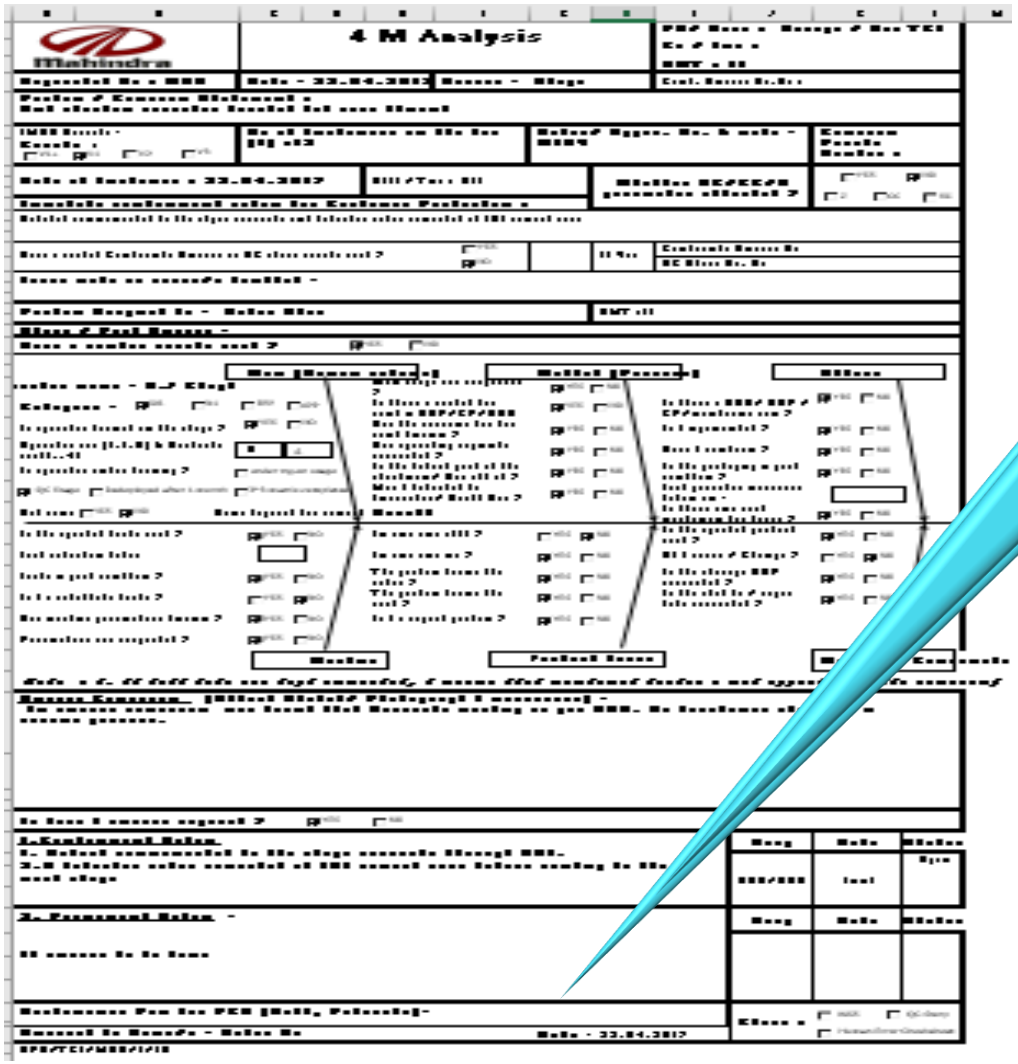
Inference – Defect rate is consistently increasing month on month (Marazzo Productions is also increasing).

3. Understand the current situation

Rocker panel scratches

Rocker panel damage

3. Understand the current situation



Inference from L0
Problem is a repeat problem
L1 analysis required

4M analysis Good and Not so good observations	
MAN	Operator dext. Level L4
Method	Main step are respected by associate.
Others	Scratches & damage generated between trim & fai.
Machine	Tools used are calibrated
Material	Handling and shelf life of part adhered

No Quick Win

Step 3. Understanding Current Situation

4M analysis Good & Not so good observations

- Associate is dexterity L4.
- Main step are respected by associate.
- Stage adherence found 100%.
- Associate is following his SOS .
- Change event was new model.
- Issue observed in both shifts on Dent and paint inspection stage.
- Scratches observed on Both side LH/RH.
- Analysis time more to identify scratches and damages concern.
- Rework time more .

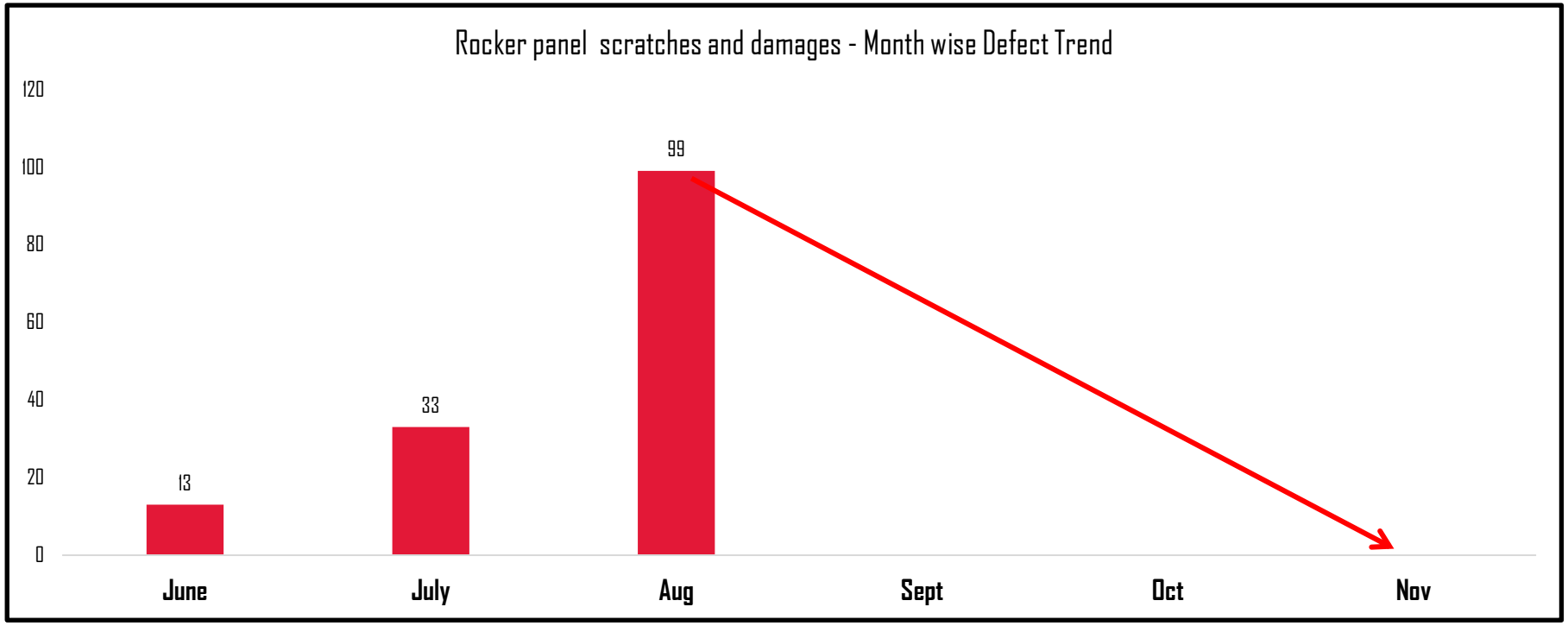
4. Select targets

Present Level:

Average 40-50 Defects/month

Target:

Elimination of the defect Rocker panel scratches and damages By NOV 2018



5. Analysis

5.1 List of Possible causes: Brainstorming

Sr. no	Possible Causes
1	New associate deployed at stage.
2	SOS Not Followed .
3	Seat mounting gun foul on rocker panel.
4	Unskilled associate for stage.
5	As it is ok concern detected.
6	Metal parts of body fouling to vehicle body while work.
7	While IP fitment , IP edges fouls to rocker panel .
8	Child part Trolleys fouls to the rocker panel .
9	Hitting of Hanger because of the uneven distance between the arms-pegs of the hanger while dropping the vehicle at drooping stage of Trim 2.
10	Damages received from body shop .
11	No gap between two trolleys at storage area .
12	While CCB fitment CCB fouls on rocker panel.
13	Rocker panel damage while seat fitment.
14	While dash panel fitment Star clips hits on rocker panel.
15	While scuff plate fitment .

Step 5. Analysis

List of Probable causes identification in Brain storming session

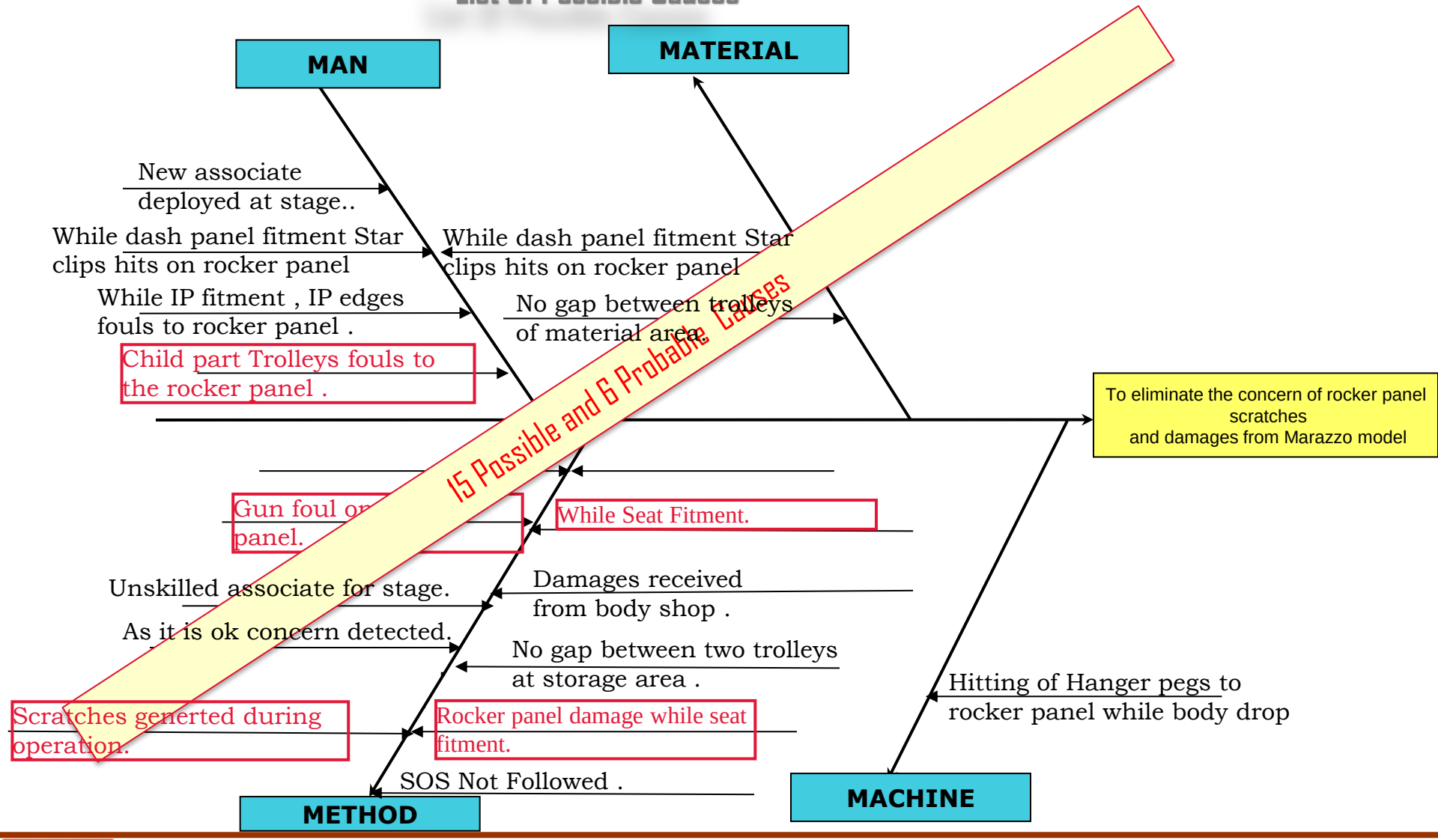
Sr. no	Possible Causes	Burhan	Ganesh	Santosh	Prasad	Nitin	Taj	Total
1	New associate deployed at stage.	1	3	1	3	1	3	12
2	SOS Not Followed .	1	3	1	1	3	1	10
3	Gun foul on rocker panel.	9	3	3	9	9	9	42
4	Unskilled associate for stage.	1	3	1	3	1	3	12
5	As it is ok concern detected.	3	1	3	1	1	1	10
6	Scratches generted during operation	9	9	3	3	9	9	42
7	While IP fitment , IP edges fouls to rocker panel .	1	3	3	1		1	9
8	Child part Trolleys fouls to the rocker panel .	3	9	9	9	3	9	42
9	Hitting of Hanger pegs to rocker panel while body drop	9	9	3	9	9	3	42
10	Damages received from body shop .	1	3	1	3	1	1	10
11	No gap between two trolleys at storage area .	1	3	3	1	1	3	12
12	While Seat Fitment.	9	1	9	3	9	9	40
13	Rocker panel damage while seat fitment.	9	3	9	9	9	3	42
14	While dash panel fitment Star clips hits on rocker panel.	1	3	3	1	3	1	12
15	While scuff plate fitment .	3	3	1	3	1	3	14

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.

Threshold Value for selecting probable cause is 40.

Step 5. Analysis

List Of Possible Causes



Step 5. Analysis

5.1 List of causes not considered as probable causes & reasons for not considering

Sr.No	Not considered Causes	Reasons
1	New associate deployed at stage.	All trained operator deployed on stage
2	SOS Not Followed .	Checked & found ok
4	Unskilled associate for stage.	All trained operator deployed on stage
5	As it is ok concern detected.	Checked & found ok
7	While IP fitment , IP edges fouls to rocker panel .	Checked & found ok
10	Damages received from body shop .	Checked & found ok
11	No gap between two trolleys at storage area .	Checked & found ok
14	While dash panel fitment Star clips hits on rocker panel.	Checked & found ok
15	While scuff plate fitment .	Checked & found ok

5.1 List of Probable causes: Brainstorming

Sr.No	Possible Causes
3	Gun foul on rocker panel.
6	Scratches generated during operation
8	Child part Trolleys fouls to the rocker panel .
9	Hitting of Hanger pegs to rocker panel while body drop
12	While Seat Fitment.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	Gun foul on rocker panel.	Visual Check .	25 vehicles checked in which 2 vehicles observed for Gun fouling to rocker panel and generated scratch.	Hypothesis valid

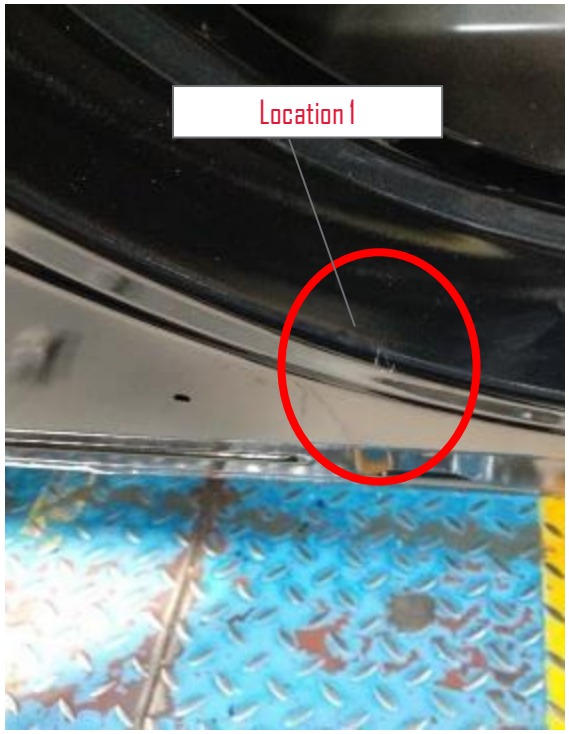
Check sheet for Gun fouling	
SR NO	Observations
1	Found ok
2	Found ok
3	Found ok
4	Found ok
5	Found ok
6	Found ok
7	Found ok
8	Found ok
9	Found ok
10	Found ok
11	Found ok
12	Found ok
13	Not ok
14	Found ok
15	Found ok
16	Found ok
17	Found ok
18	Found ok
19	Found ok
20	Found ok
21	Found ok
22	Not ok
23	Found ok
24	Found ok
25	Found ok

While doing tightening operator tights seat from outside , in which gun fouls to rocker panel.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
8	Scratches generated during operation	Audit done for 5days .In which 2 type of scratches observed on rocker panel at different location .	daily 2 to 3 vehicles reported for scratches on rocker panel for same phenomenon of scratch.	Hypothesis valid



Check sheet for Gun fouling		
SR NO	Location 1	Location 2
1	Found ok	Found ok
2	Found ok	Found ok
3	Found ok	Not ok
4	Found ok	Found ok
5	Not ok	Found ok
6	Found ok	Found ok
7	Found ok	Found ok
8	Found ok	Found ok
9	Found ok	Found ok
10	Found ok	Not ok
11	Found ok	Found ok
12	Found ok	Found ok
13	Found ok	Found ok
14	Found ok	Found ok
15	Found ok	Found ok
16	Found ok	Found ok
17	Not ok	Found ok
18	Found ok	Found ok
19	Found ok	Found ok
20	Found ok	Not ok
21	Found ok	Found ok
22	Found ok	Found ok
23	Found ok	Found ok
24	Found ok	Found ok
25	Found ok	Found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
8	Child part Trolleys fouls to the rocker panel .	Visual inspection done.	10 vehicles checked in which no suspected cause and no damage observed on rocker panel.	Hypothesis invalid

Check sheet for Trolley fouling.	
SR NO	Observations
1	Found ok
2	Found ok
3	Found ok
4	Found ok
5	Found ok
6	Found ok
7	Found ok
8	Found ok
9	Found ok
10	Found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
13	Hitting of Hanger pegs to rocker panel while body drop.	Visual check.	30 vehicles checked at body drop , no fouling observed on any hanger.	Hypothesis Invalid

Check sheet for Hanger fouling	
SR NO	Observations
1	Found ok
2	Found ok
3	Found ok
4	Found ok
5	Found ok
6	Found ok
7	Found ok
8	Found ok
9	Found ok
10	Found ok
11	Found ok
12	Found ok
13	Found ok
14	Found ok
15	Found ok
16	Found ok
17	Found ok
18	Found ok
19	Found ok
20	Found ok
21	Found ok
22	Found ok
23	Found ok
24	Found ok
25	Found ok
26	Found ok
27	Found ok
28	Found ok
29	Found ok
30	Found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
12	While Seat Fitment .	Visual inspection at Seat Fitment stage.	30 vehicles were observed in shift, in which 3 vehicles found defective.	Hypothesis valid

Check sheet for Seat Fitment	
SR NO	Observations
1	Found ok
2	Found ok
3	Found ok
4	Found ok
5	Found ok
6	Found ok
7	Found ok
8	Found ok
9	Found ok
10	not ok
11	Found ok
12	Found ok
13	Found ok
14	Found ok
15	Found ok
16	not ok
17	Found ok
18	not ok
19	Found ok
20	Found ok
21	Found ok
22	Found ok
23	Found ok
24	Found ok
25	Found ok
26	Found ok
27	Found ok
28	Found ok
29	Found ok
30	Found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Gun foul on rocker panel.	Visual Check .	25 [illegible] in which 2 vehicles [illegible] gun fouling to rocker panel and [illegible]rated scratch.	Hypothesis valid
2	Scratches generated during operation	Audit done for [illegible] which 2 t[illegible] [illegible]tches obse[illegible] [illegible]cker panel [illegible]t location .	daily 2 to 3 vehicles reported for scratches on rocker panel for same phenomenon of scratch.	Hypothesis valid
3	While Seat Fitm[illegible]	Visual inspection at Seat Fitment stage.	30 vehicles were observed in shift, in which 3 vehicles found defective.	Hypothesis valid

3 Valid Causes

5.3 Identify the root causes:

WHY - WHY Analysis

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Gun foul on rocker panel.	Seat bolt tightening done from outside of the vehicle & during tightening battery bottom fouls to rocker area	Tightening outside or inside is not defined	This failure mode not identified	Change the tightening position
2	Scratches generated during operation	While IP fitment Shoe lace rivet fouls to rocker panel causing scratch at location 1.	While water shield fitment one associate having artificial leg , which fouls to rocker panel causing scratch.	No protection on rocker panel	Add protection
3	While Seat Fitment .	As seat is bulky part hence for insertion manipulator required.	While insertion of seat in vehicle Rocker panel is blind for operator , which is generating damage	No protection on rocker panel	Add protection

6. Implement the corrective measures

6.1 Actions Decided

No	Root Cause	Action plan	Target date	Responsibility	Status
1	This failure mode not identified	Change the tightening position	08.09.18	Burhan Merchant	Closed
2	No protection on rocker panel	Add protection	15.9.18	Burhan Merchant	Closed
3	No protection on rocker panel	Add protection	10.9.18	Ganesh Bendale	Closed

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	This failure mode not identified – Battery fouling	Tightening of seat bolt from inside of vehicle is now change to inside.	8.9.18	Burhan Merchant	Closed

Before

After

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	No protection on rocker panel	3M thickness Strip is used to stick on Rocker Panel to avoid Scratches	15.9.18	Burhan Merchant	Closed

Before

After

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
2	No protection on rocker panel - To avoid seat leg fouling	Magnetic FRP Cover fixture Started before insertion of Seat to Avoid Scratches	10.9.18	Burhan Merchant	Closed

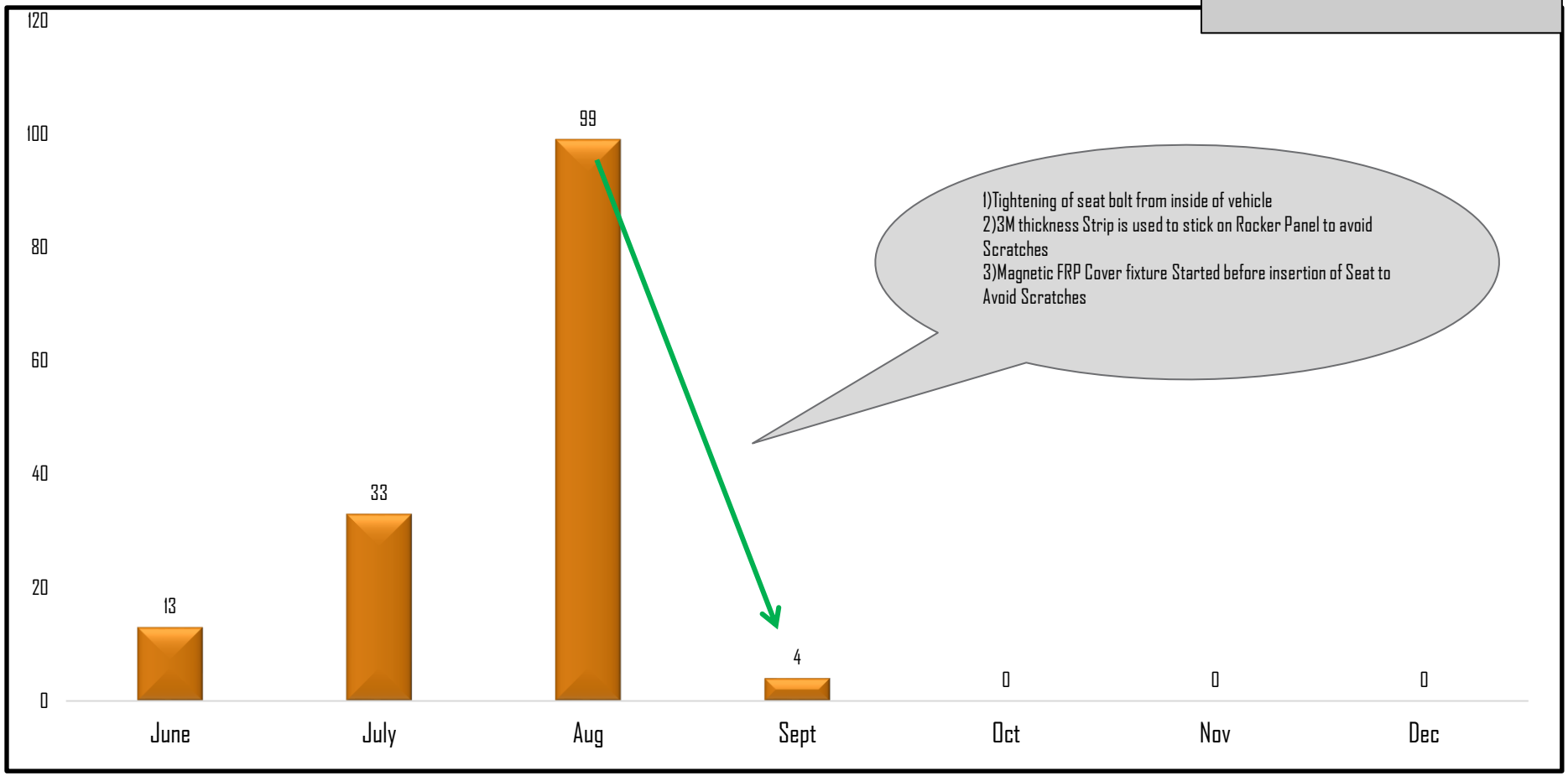
Before

After

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend

X Axis- Month
Y Axis- Rocker panel scratches and damages incidences in marazzo model.



Inference - Concern is eliminated after all 3 actions implemented.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

No	Root Cause	Action plan	Target date	Responsibility	Status
1	Seat Bolt tightening done from outside of vehicle.	Tightening of seat bolt from inside of vehicle	8.9.18	Burhan Merchant	Closed
2	No protection on rocker panel	3M thickness Strip is used to stick on Rocker Panel to avoid Scratches	15.9.18	Burhan Merchant	Closed
3	While insertion of seat in vehicle Rocker panel is blind for operator , which may generate damage.	Magnetic FRP Cover fixture Started before insertion of Seat to Avoid Scratches	10.9.18	Ganesh Bendale	Closed

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	SOS UPDATION SEAT BOLT FITMENT.	8.9.18	Ganesh Bendale	Closed
2	PCRN FOR 3M TAPE	15.9.18	Nitin Aher	Closed
3	SOS UPDATION FOR FRP WHILE SEAT LOADING	10.9.18	Ganesh Bendale	Closed
4	FRP FIXTURE LIST UPDATION.	10.9.18	Merchant Burhan	Closed
5	QA matrix	15.9.18	Merchant Burhan.	Closed
6	Process audit check sheet	15.9.18	Merchant Burhan	Closed

Check & confirm the results & effects (Cont..)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) STR improved from 60 to 62 %
2) COPQ (Material cost+ Labour cost+ Rework cost) =(Apprx. Rs. 30154. Rs. Per annum)is Saved

In tangible Benefits :

1) Improved Customer Satisfaction.
2) Waiting time of vehicle reduced.

From: DHANOKAR RAJESH
Sent: Tuesday, December 18, 2018 4:09 PM
To: MERCHANT BURHAN--NASHIK
Subject: Re:MYB F19 COPQ Approval

ckay Approved

Sent from my Redmi Note 4X
On MERCHANT BURHAN--NASHIK <MERCHANT.BURHAN2@mahindra.com>, 18-Dec-2018 4:01 PM wrote:

Dear Sir,

Please approved.

Rework Cost estimation		
Description	Units	
Rework time (Approx)	Min	60
Avg Concerns/Day	Nos	2
Total Required Rework time per day	Min	120
Effective working time	Min	465
Operator required /day	nos	0.26
Approx Operator cost/ day	Rs	325
Operator cost /concern	Rs	83.87
Avg no. of working days	days	24
Operator cost /month	Rs	2012.90
Part reject/month	Nos	500.00
Part Cost	Rs	1.00
Operator Cost/year	**Rs**	**30154.84**

8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

SN	Document	Target date	Responsibility	Status
1	SOS UPDATION SEAT BOLT FITMENT.	8.9.18	Merchant Burhan	Closed
2	PCRN FOR 3M TAPE	15.9.18	Merchant Burhan	Closed
3	SOS UPDATION FOR FRP WHILE SEAT LOADING	10.9.18	Ganesh Bendale	Closed
4	FRP FIXTURE LIST UPDATION.	10.9.18	Ganesh Bendale	Closed
5	QA matrix	15.9.18	Merchant Burhan	Closed

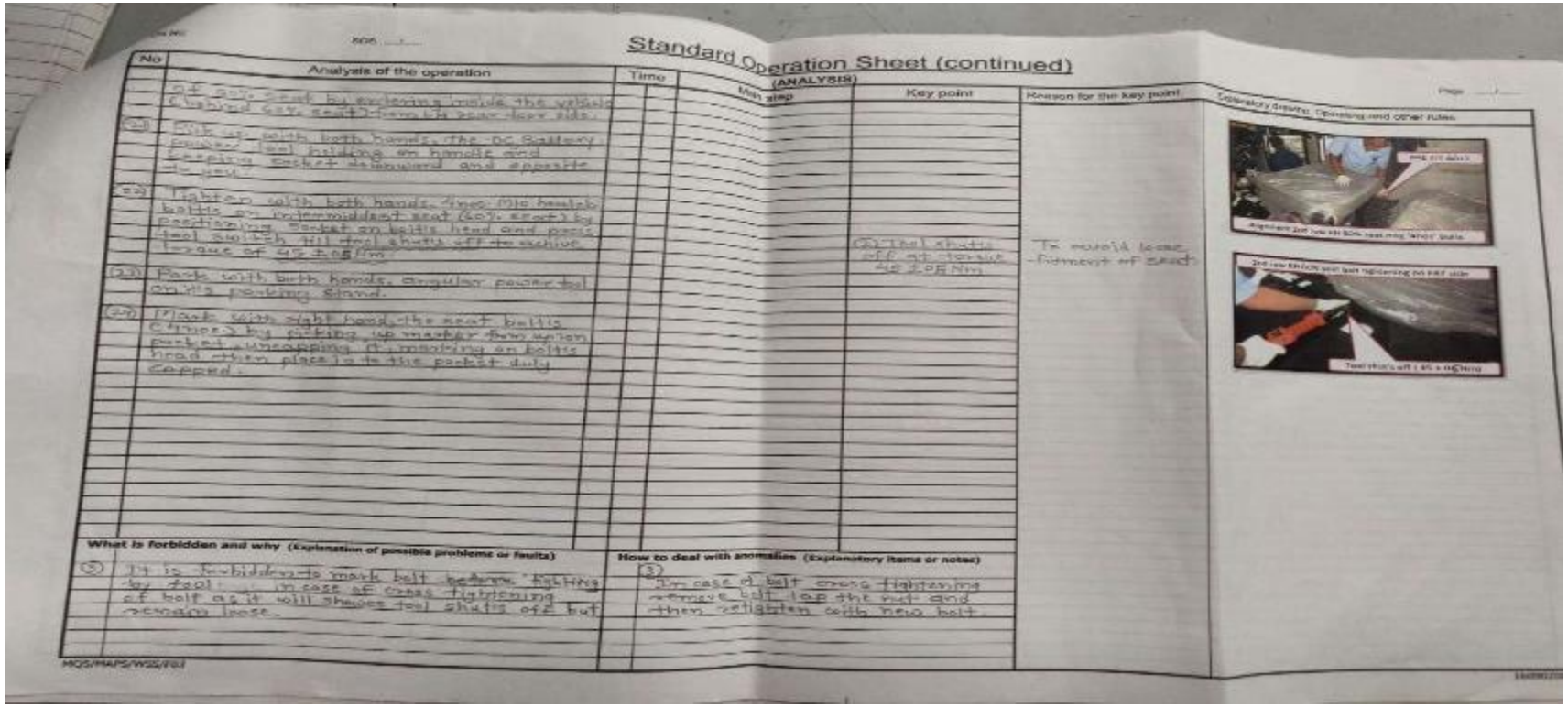
8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	OPL given for On job training given to all stage.	15.9.18	Merchant Burhan	Closed

8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

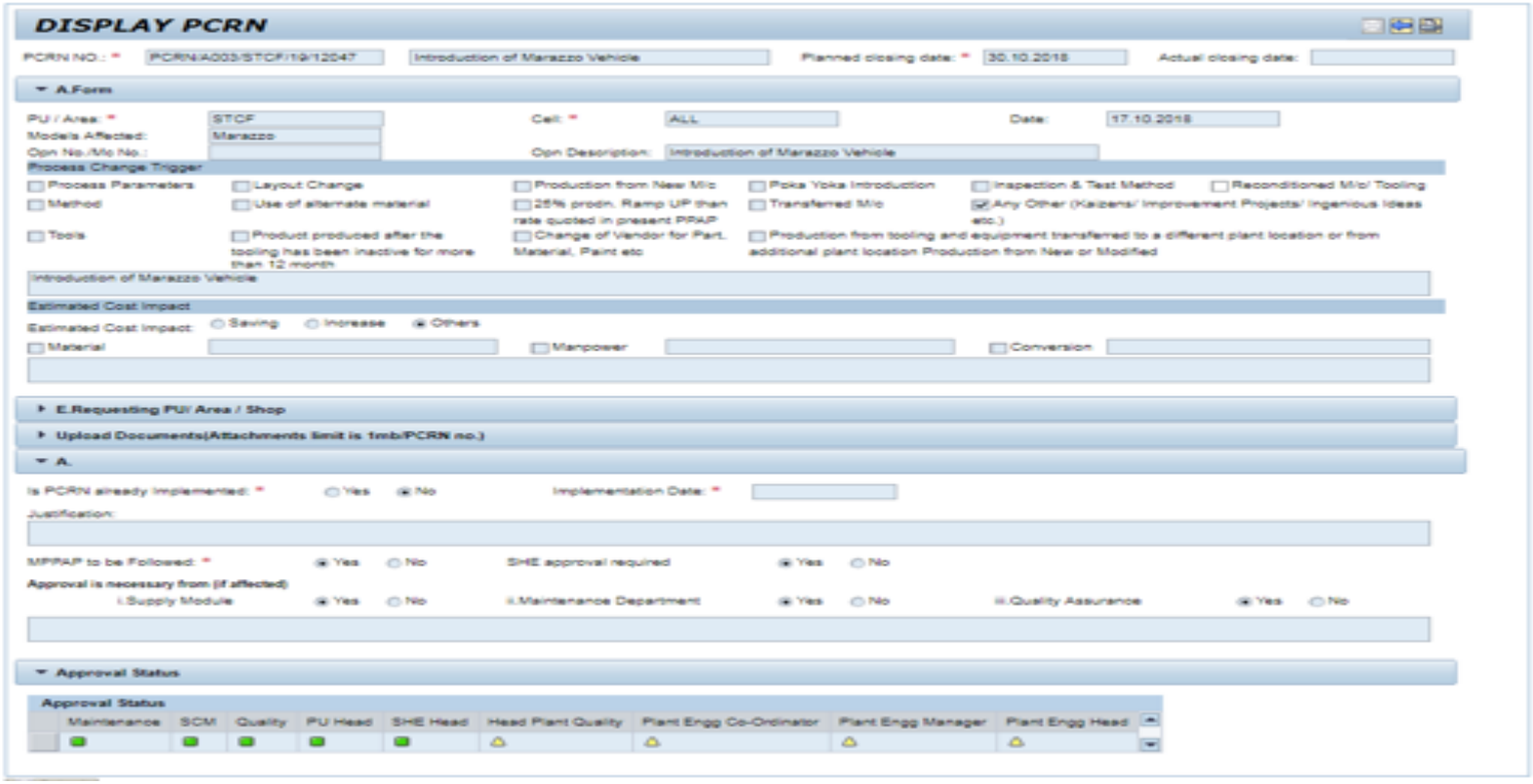
SN	Document	Target date	Responsibility	Status
1	SOS UPDATION SEAT BOLT FITMENT	8.9.18	MERCHANT BURHAN	Closed



8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

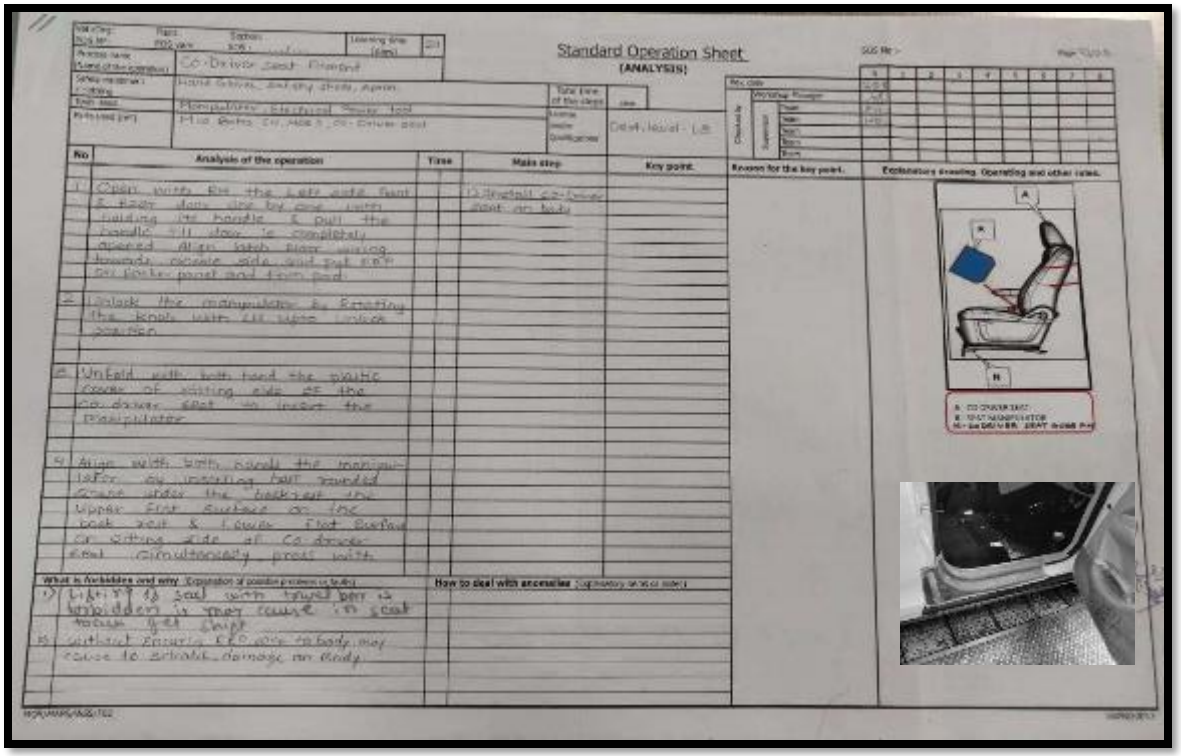
SN	Document	Target date	Responsibility	Status
2	PCRN FOR 3M TAPE	15.9.18	MERCHANT BURHAN	Closed



8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

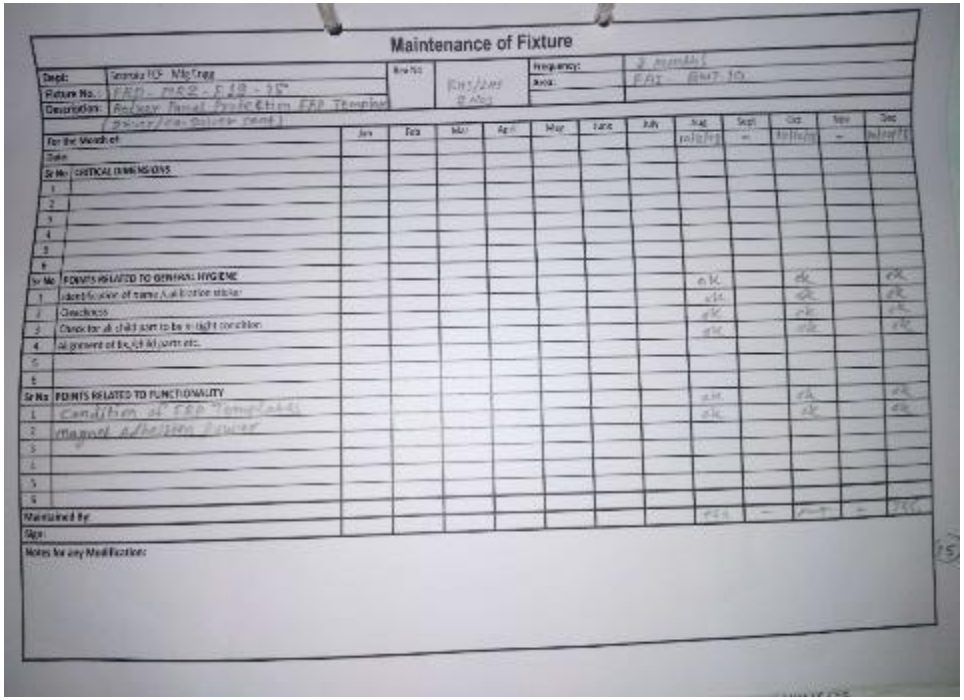
SN	Document	Target date	Responsibility	Status
3	SOS UPDATION FOR FRP WHILE SEAT LOADING	10.9.18	MERCHANT BURHAN	Closed

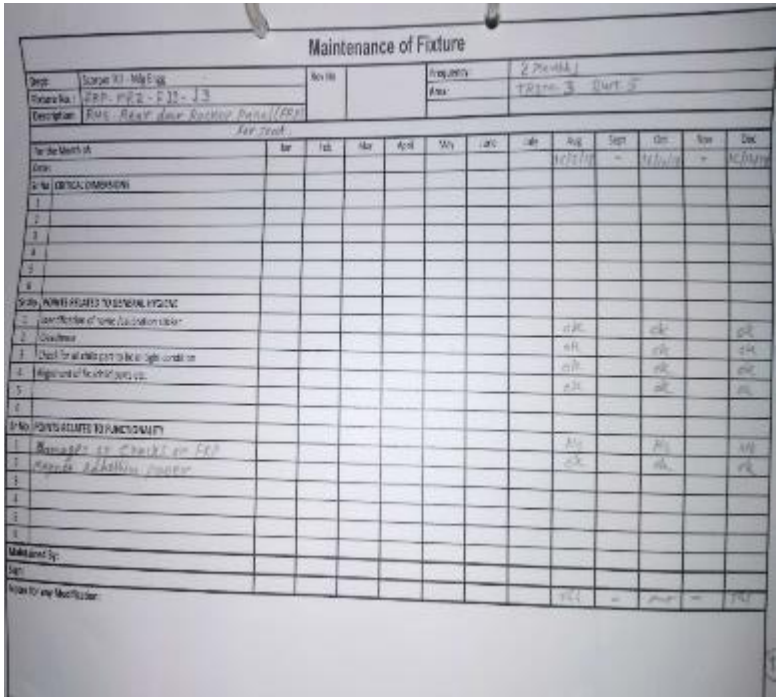


8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

SN	Document	Target date	Responsibility	Status
4	FRP FIXTURE LIST UPDATION.	15.9.18	MERCHANT BURHAN	Closed





8. Standardization

8.1 Updation of documents: e.g. Part Drawing,EN,SOS,PFMEA,Check sheet.

SN	Document	Target date	Responsibility	Status
5	QA matrix	15.9.18	MERCHANT BURHAN	Closed

QA MATRIX

PU NAME	Scorpio/Xylo
DEPT NAME	TCF
SECTION NAME	Trim 3

INITIALIZATION DATE	01.04.2016			
SIGNATURES				
Dept Head	R P Narkar			
Module Manager	S B Chikhale			
Section Manager	1	2	3	4

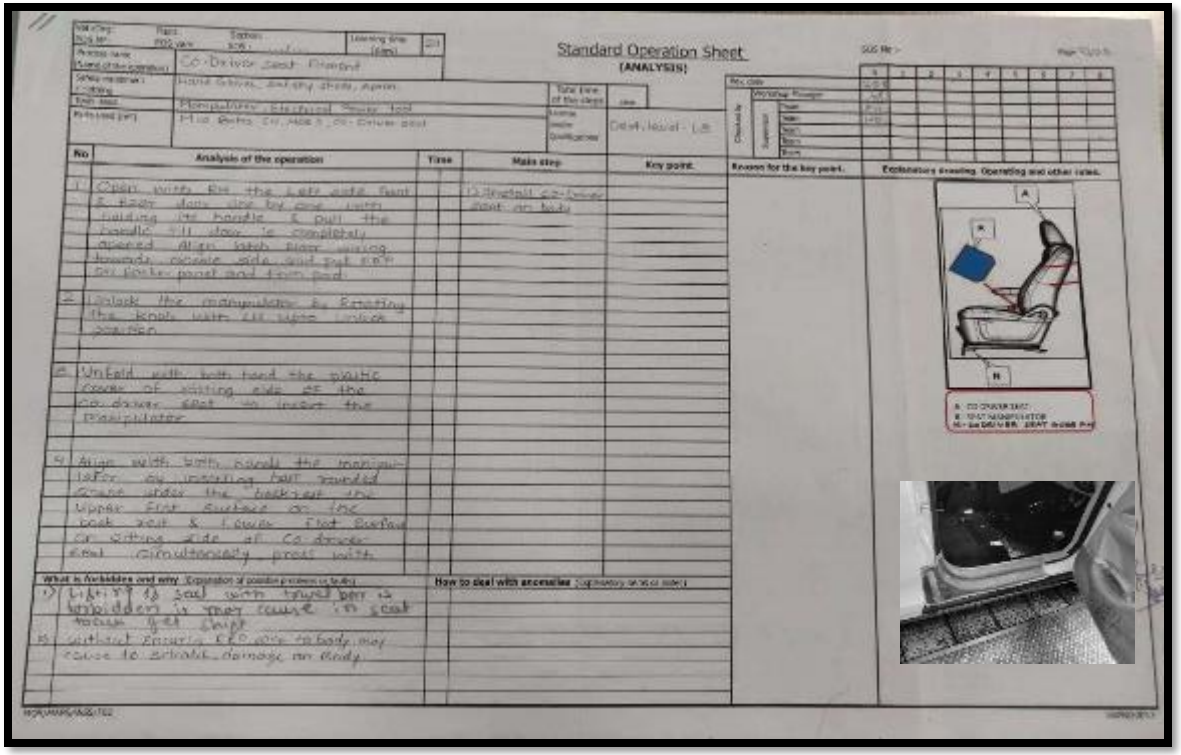
Global Quality Guarantee level.	
Section	100%
Plant	100%
Revision date	15.9.18

Workstation No.	Process description (PDS)	Date	Failure modes	Severity	Fail occurrence	Fail detection	Inspection process: Section			Inspection process: Outside section			Guarantee value				Faults recorded			Quality guarantee level				Comments
							Scorpio/Xylo	pass Trim 3	Key off	Roller	Door	dashboard inspection	WHOP	Workstation	Section	Outside section	PLANT	Section (IM)	Outside section (IM)	CUSTOMERS (IM)	WORKSTATION	Section	Outside section	PLANT	
1	Seat Fitment		Roller panel damage	9	2	3	⊙ 3					⊙ 3	⊙ 3	3	3	3	6	0	0	0	ok	OK	OK	OK	
2	Seat Bolt tightening		Roller panel damage	9	2	3	⊙ 3					⊙ 3	⊙ 3	3	3	3	6	0	0	0	ok	OK	ok	OK	

8. Standardization

8.2 Training

5	On job training given to all stage associates for use of fixture while seat insertion.	15.9.18	MERCHANT BURHAN	CLOSED

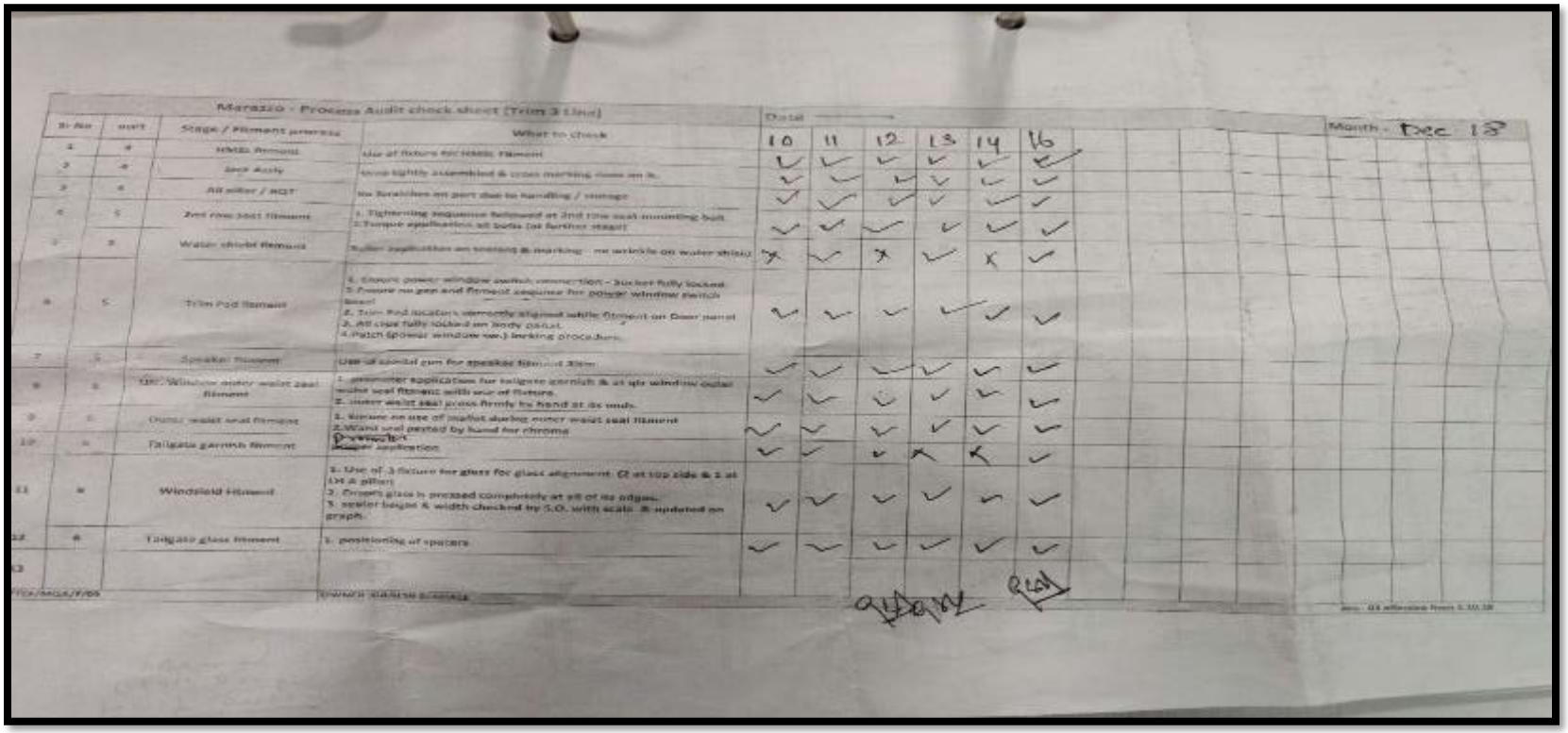


8. Standardization

8.3 Additional Audit / Check

5	Additional check Point in Process audit check sheet.	15.9.18	MERCHANT BURHAN	CLOSED

Point Added in S.O. Audit check sheet .



9. Summaries & Plan future actions

9.1 Conclusion:

Rocker panel scratches concern is eliminated in Marazzo after

- Tightening of seat bolt from inside of vehicle
- 3M thickness Strip is used to stick on Rocker Panel to avoid Scratches
- Magnetic FRP Cover fixture Started before insertion of Seat to Avoid Scratches

9.2 Horizontal Deployment:

All U322 and U325 model From VP gateway.

9.3 Lessons Learnt:

Things Gone Wrong :

Seat insertion operation was probable source for Scratches / Damages but no proactive actions were identified to protect.

Things Gone Right :

Process walk through with CFT from body to TCF given us lot of inputs – like – One Associate was having Artificial leg which was generating the scratches.

Thank You.....